

Chapter

Work, Power and Energy

1. For burning of coal in a thermoelectric station, the energy conversion taking place is:

- (a) chemical to heat to mechanical
- (b) chemical to heat to mechanical to electrical
- (c) chemical to heat to light
- (d) heat to chemical to mechanical

SQP 2025

2. The kinetic energy of a given body depends on the:

- (a) position
- (b) centre of gravity of the body
- (c) momentum
- (d) displacement

SQP 2025

3. If the mass as well as the velocity of a body is doubled then the kinetic energy of the body (is doubled/becomes eight times / becomes four times) the initial kinetic energy.

SQP 2025

4. Unit of power used in mechanical engineering is (watt / horse power / erg per second)

SQP 2025

5. Draw a graph of potential energy vs height for a body thrown vertically upwards. [Assume no friction is present.]

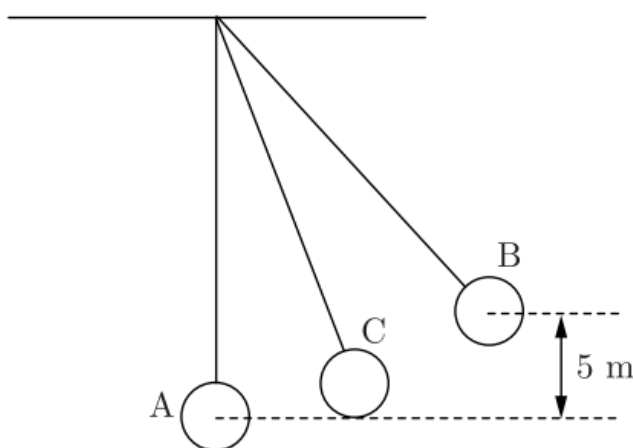
SQP 2025

6. For each of the following scenarios, state whether the work done by gravity is positive, negative, or zero.

- (a) a person walks on a levelled road.
- (b) a person climbs a ladder.
- (c) a car in neutral gear is coming down the slope.

SQP 2025

7. The figure below shows a simple pendulum of mass 200 g. It is displaced from the mean position A to the extreme position B. The potential energy at the position A is zero. At the position B the pendulum bob is raised by 5 m.



- (a) What is the potential energy of the pendulum at the position B?
 - (b) What is the total mechanical energy at point C?
 - (c) What is the speed of the bob at the position A when released from B?
- (Take $g = 10 \text{ ms}^{-2}$ and given that there is no loss of energy.)

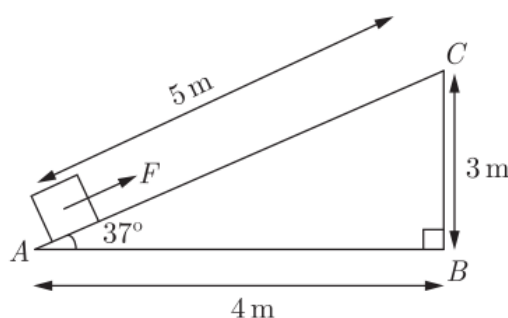
SQP 2025

8. When a bell fixed on a cycle rings, then the energy conversion that takes place is

- (a) gravitational potential energy to sound energy
- (b) kinetic energy to sound energy
- (c) sound energy to electrical energy
- (d) sound energy to mechanical energy

MAIN 2024

9. A force 'F' moves a load from A to C as shown in the figure below. For the calculation of the work done, which of these lengths would you use as the displacement?



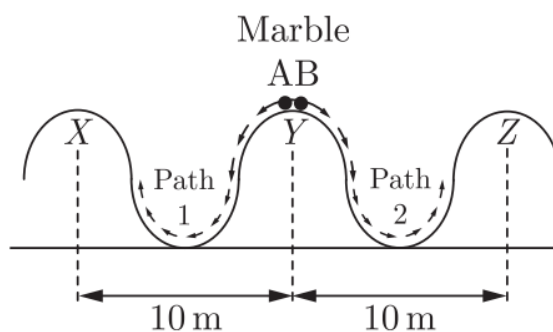
- (a) 3 m (b) 4 m
(c) 5 m (d) 7 m

MAIN 2024

10. Why does it become pleasantly warm when the lakes start freezing?

MAIN 2024

11. Two identical marbles A and B are rolled down along Path 1 and Path 2 respectively. Path 1 is frictionless and Path 2 is rough.



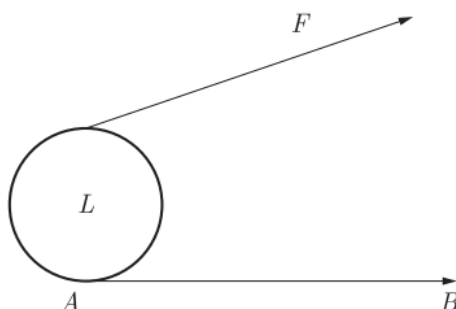
- (a) Which marble will surely reach the next peak?
(b) Along which path/s the mechanical energy will be conserved?
(c) Along which path/s is the law of conservation of energy obeyed?

MAIN 2024

12. Sumit does 600 J of work in 10 min and Amit does 300 J of work in 20 min. Calculate the ratio of the powers delivered by them.

MAIN 2024

13. A boy drags a load L along horizontal plane AB by applying a force F . The boy does



- (a) some positive work
- (b) no work
- (c) negative work
- (d) none of these

SQP 2023

14. State the transformation of energy taking place in a:

- (i) Loudspeaker
- (ii) Washing machine

SQP 2023

15. How can the work done be measured when force is applied at an angle to the direction of displacement?

SQP 2023

16. How do you measure the work when the force is applied at an angle?

SQP 2023

17. If the speed of a car is halved, by which factor does its kinetic energy change?

MAIN 2023

18. What is the SI unit of power? How is it related to horse power?

MAIN 2023

19. Calculate the power of a pump which can lift 400 kg of water to store it in a water tank at a height of 19 m in 40 s. (Take $g = 10 \text{ ms}^{-2}$)

MAIN 2023

20. Work done = force ×

- (a) displacement
- (b) distance
- (c) time
- (d) velocity

COMP 2023

21. In a hydroelectric power station, 1000 kg of water is allowed to drop a height of 100 m in 1 s. If the conversion of potential energy to electric energy is 60%, calculate the output power.

[Take $g = 10 \text{ ms}^{-2}$]

- (a) $4 \times 10^5 \text{ W}$
- (b) $1.5 \times 10^5 \text{ W}$
- (c) $8 \times 10^5 \text{ W}$
- (d) $6 \times 10^5 \text{ W}$

COMP 2023

22. What energy conversions take place in the following when they are working:

(a) electric toaster, and (b) microphone?

COMP 2023

23. A machine which is 75% efficient, uses 1.2 J of energy in lifting up a 2.0 kg mass through a certain distance. This mass is then allowed to fall through that distance. What will be its velocity at the end of its fall?

COMP 2023

24. Power is the product of

- (a) force and time
- (b) force and acceleration
- (c) force and displacement
- (d) force and velocity

SQP 2022

25. Energy is

- (a) capacity to produce power
- (b) capacity to do work
- (c) capacity to do work per unit time
- (d) the capacity to use power

MAIN 2022

26. How is nuclear energy generated?

COMP 2022

27. When an arrow is shot from its bow, it has kinetic energy. From where does it get the kinetic energy?

COMP 2022

28. State the amount of work done by an object when it moves in a circular path for one complete rotation. Give a reason to justify your answer.

COMP 2022

29. What force must be applied to a body through a distance of 10 m, such that it does a work of 4000 J. If the mass of the body is 20 kg, what is the acceleration of the body ?

(a) 20 ms^{-2}

(b) 10 ms^{-2}

(c) 40 ms^{-2}

(d) 30 ms^{-2}

SQP 2021

30. A catapult throws a stone of mass 0.10 kg with a velocity of 30 ms^{-1} . If 25% of the potential energy of the elastic band is wasted during transmission, find the magnitude of the potential energy.

(a) 78 J

(b) 60 J

(c) 375 J

(d) 215 J

SQP 2021

31. What is dissipation of energy ?

SQP 2021

32. Define 'joule', write SI unit of work and establish a relationship between the SI and CGS unit of work.

SQP 2021

33. A spring is kept compressed by a small trolley of mass 500 g lying on a smooth horizontal surface. When the trolley is released, it is found to move with a speed of 2 ms^{-1} . What potential energy did the spring possess when compressed ?

SQP 2021

34. What is the amount of force required to be applied on a body so as to displace it through 20 m performing 8000 J of work?

- (a) 400 N (b) 400 J
(c) 40 N (d) 40 J

MAIN 2021

35. Define a kilowatt hour. How is it related to the joule?

MAIN 2021

36. A body of mass 4 kg initially at rest is subjected to a force of 16 N. What is the kinetic energy acquired by the body at the end of 10 s?

MAIN 2021

37. A coolie raises a load upwards against the force of gravity, then the work done by the load is

- (a) zero (b) positive work
(c) negative work (d) None of these

SEM-I 2021

38. 1 J = erg

- (a) 10^9 (b) 10^7
(c) 10^5 (d) 10^6

SEM-I 2021

39. A light body A and a heavy body B have the same momentum.

SEM-I 2021

(i) Choose a correct statement from the given options.

- (a) Kinetic energy of body A and body B will be the same.
(b) Kinetic energy of body A is greater than kinetic energy of body B .
(c) Kinetic energy of body B is greater than kinetic energy of body A .
(d) Unless we know the velocity, we cannot find which body has greater kinetic energy.

(ii) If the ratio of kinetic energies of A and B is 5 : 2, then which one of the following gives the mass ratio of the bodies respectively?

- (a) 5 : 2 (b) 25 : 4
(c) 2 : 5 (d) 4 : 24

40. The energy change during photosynthesis in plants is

- (a) heat to chemical (b) light to chemical
(c) chemical to light (d) chemical to heat

SEM-I 2021

41. A body of mass 200 g falls freely from a height of 15 m. (Take, $g = 10 \text{ ms}^{-2}$)

SEM-I 2021

(i) When the body reaches 10 m above the ground, its potential energy will be

- (a) 20000 J (b) 10 J
(c) 10000 J (d) 20 J

(ii) The total mechanical energy it will possess, when it is just about to strike the ground is

- (a) 30000 J (b) 20000 J
(c) 30 J (d) 20 J

(iii) The velocity in ms^{-1} with which the body will hit the ground is

- (a) 30 (b) 10
(c) $10\sqrt{3}$ (d) $10\sqrt{2}$

42. What do you understand by kinetic energy of a body? Give its SI unit. Also give two examples from everyday life.

COMP 2021

43. Rajan exerts a force of 150 N in pulling a cart at a constant speed of 10 m/s. Calculate the power exerted.

COMP 2021

Solution

1.

Ans: (b) chemical $\rightarrow$ heat $\rightarrow$ mechanical $\rightarrow$ electrical

Explanation:

In a thermoelectric (thermal power) station:

1. **Chemical energy** in coal is released by burning.
2. This converts into **heat energy** (steam is produced).
3. High-pressure steam drives a turbine $\rightarrow$ **mechanical energy**.
4. The turbine rotates a generator $\rightarrow$ **electrical energy**.

So, the energy conversion is: **chemical** $\rightarrow$ **heat** $\rightarrow$ **mechanical** $\rightarrow$ **electrical**.

2.

Ans: (c) momentum

Explanation:

Kinetic energy K of a body can be written in terms of momentum p as:

$$K = \frac{p^2}{2m}$$

Here m is the mass (constant for a given body), so K depends directly on p^2 .

Thus, kinetic energy of a given body depends on its momentum, not on position, centre of gravity, or displacement.

3.

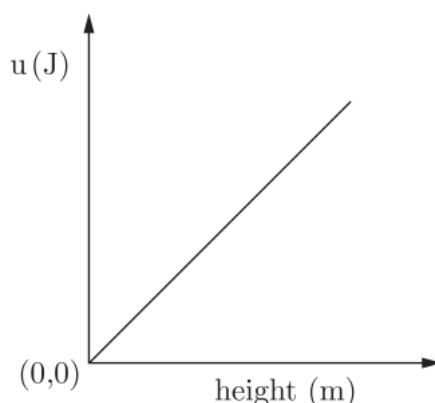
Ans: becomes eight times

4.

Ans: horse power

5.

Ans:



6.

Ans: (a) Zero work is done.

Gravity acts vertically downwards, while the person's displacement is horizontal, so the angle between them is 90° . Hence work done by gravity is zero.

(b) Negative work is done.

When a person climbs up, displacement is upward but gravity acts downward, opposite to the direction of motion. So, the work done by gravity is negative.

(c) Positive work is done.

For a car rolling down a slope in neutral, displacement is downward along the slope and the component of gravitational force is in the same direction. Hence gravity does positive work.

7.

Ans: (a) Potential energy at position B:

$$U = mgh = 0.2 \times 10 \times 5 = 10 \text{ J}$$

(b) Total mechanical energy at point C = **10 J**

(Conservation of mechanical energy)

(c) Using the principle of conservation of energy,

Potential energy at B = Kinetic energy at A

$$10 = \frac{1}{2} \times 0.2 \times v^2$$

$$v^2 = 100$$

$$v = 10 \text{ m/s}$$

8.

Ans: (b) kinetic energy to sound energy

Explanation:

When a bell on a cycle is struck, the **mechanical (kinetic) energy** from the finger or striker causes the bell to **vibrate**. These vibrations produce **sound waves**.

Hence, the energy conversion taking place is from **kinetic energy to sound energy**.

9.

Ans: (c) 5 m

Explanation:

Work done $W = F \times$ displacement in the direction of the force.

Here, the load is moved along the **inclined plane from A to C**, which is **5 m**, in the same direction as the force F .

So, displacement to be used is **5 m**.

10.

Ans: When a lake begins to freeze, it releases **latent heat** into the surroundings.

This heat warms the air around the lake, making the atmosphere feel pleasantly warm.

11.

Ans: (a) Marble **A** will surely reach the next peak.

Explanation:

Path 1 is frictionless, so no energy is lost.

(b) Mechanical energy will be conserved **along both paths**.

Explanation:

Energy changes between potential and kinetic, but is still accounted for.

(c) The law of conservation of energy is obeyed **along both paths**.

12.

Ans:

Power of Sumit,

$$P_{\text{Sumit}} = \frac{W}{t} = \frac{600 \text{ J}}{10 \text{ min}} = 60 \text{ (J/min)}$$

Power of Amit,

$$P_{\text{Amit}} = \frac{W}{t} = \frac{300 \text{ J}}{20 \text{ min}} = 15 \text{ (J/min)}$$

Ratio of their powers,

$$\frac{P_{\text{Sumit}}}{P_{\text{Amit}}} = \frac{60}{15} = \frac{4}{1}$$

So, the ratio of powers delivered by Sumit and Amit is **4 : 1**.

13.

Ans: (a) some positive work

Explanation:

The boy applies a force $\vec{F}$ at an angle to move the load L along the horizontal plane AB .

Since the force has a **component in the direction of motion**, work done by the boy is **positive**.

Hence, the boy does **some positive work**.

14.

Ans: (i) Loudspeaker: Electrical energy $\rightarrow$ Sound energy

(ii) Washing machine: Electrical energy $\rightarrow$ Mechanical energy (drum motion) + Sound energy

15.

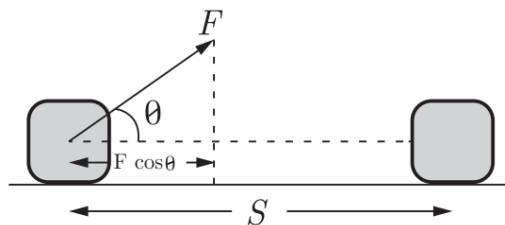
Ans: When a force F is applied at an angle θ to the direction of displacement d , the work done is given by

$$W = Fd \cos \theta.$$

16.

Ans: If a force is applied at an angle θ to the horizontal and the body moves through a displacement s , only the horizontal component of the force ($F \cos \theta$) produces work.

So, the work done is:



$$W = (F \cos \theta)s$$

$$W = Fs \cos \theta$$

Thus, work is measured using the component of the force in the direction of displacement.

17.

Ans: The kinetic energy becomes **one-fourth** of its original value.

Explanation:

Kinetic energy is given by

$$\text{KE} = \frac{1}{2}mv^2$$

If the speed is halved, new speed $v_1 = \frac{v}{2}$.

New kinetic energy:

$$\text{KE}_1 = \frac{1}{2}m \left(\frac{v}{2} \right)^2 = \frac{1}{2}m \cdot \frac{v^2}{4} = \frac{1}{4} \left(\frac{1}{2}mv^2 \right) = \frac{1}{4}\text{KE}$$

So, when speed is halved, kinetic energy becomes **one-fourth** (reduced by a factor of 4).

18.

Ans: The SI unit of power is **joule per second (J/s)**, which is also called **watt (W)** in honour of James Watt.

$$1 \text{ watt} = 1 \text{ joule/second} = 1\text{J/s}$$

The relation with horse power is:

$$1 \text{ horsepower} = 746 \text{ watt}$$

19.

Ans:

Given:

$$m = 400 \text{ kg}, h = 19 \text{ m}, g = 10 \text{ m s}^{-2}, t = 40 \text{ s}$$

Work done in lifting the water,

$$W = mgh = 400 \times 10 \times 19 = 76,000 \text{ J}$$

Power of the pump,

$$P = \frac{W}{t} = \frac{76,000}{40} = 1,900 \text{ W}$$

So, the power of the pump is **1900 W**.

20.

Ans: (a) displacement

Explanation:

Work is defined as the product of the **force** applied on a body and the **displacement** of the body in the direction of the force.

$$\text{Work done} = \text{Force} \times \text{Displacement}$$

Thus, option **(a)** is correct.

21.

Ans: (d) $6 \times 10^5 \text{ W}$

Explanation:

Potential energy of water per second:

$$PE = mgh = 1000 \times 10 \times 100 = 10^6 \text{ J}$$

If only **60%** of this energy is converted to electrical energy:

$$\text{Output energy per second} = 0.6 \times 10^6 = 6 \times 10^5 \text{ J}$$

Since this is per second,

$$\text{Output power} = 6 \times 10^5 \text{ W}$$

Hence, the correct option is **(d)**.

22.

Ans: (a) Electrical energy is converted into heat energy.

(b) Sound energy is converted into electrical energy.

23.

Ans: Velocity of the mass at the end of its fall = **3 m s⁻¹**

Explanation:

Efficiency of the machine

$$\eta = \frac{\text{Useful work done}}{\text{Total work done}} = 75\% = 0.75$$

Total work done by the machine = 12 J

So,

$$\text{Useful work done} = 0.75 \times 12 = 9 \text{ J}$$

When the mass is allowed to fall back, this 9 J of useful work becomes its kinetic energy:

$$\text{K.E.} = \frac{1}{2}mv^2$$

Given $m = 2.0 \text{ kg}$,

$$9 = \frac{1}{2} \times 2 \times v^2 \Rightarrow 9 = v^2 \Rightarrow v = 3 \text{ m s}^{-1}$$

So, the mass reaches the bottom with a speed of **3 m s⁻¹**.

24.

Ans: (d) force and velocity

Explanation:

Power is the rate of doing work.

Work $W = \text{Force} \times \text{Displacement}$, and velocity $v = \frac{\text{Displacement}}{\text{Time}}$.

So,

$$P = \frac{W}{t} = \frac{F \times s}{t} = F \times \frac{s}{t} = F \times v$$

Hence, power = **force × velocity**.

25.

Ans: (b) capacity to do work

Explanation:

Energy is defined as the **capacity or ability of a body to do work**. The more energy an object has, the more work it can perform.

Hence, **option (b)** is correct.

26.

Ans: Nuclear energy is produced during **nuclear fission**.

Explanation:

When the nucleus of a heavy atom like **uranium** or **plutonium** is bombarded with slow (low-energy) neutrons, it **splits into two lighter nuclei**. These splitting releases a **large amount of energy**, which is known as **nuclear energy**.

27.

Ans: A stretched bow has **potential energy** because its shape is changed when it is drawn back.

When the bow is released, this **potential energy is converted into kinetic energy**, which is transferred to the arrow. Thus, the arrow gets its kinetic energy from the **stored potential energy of the stretched bow**.

28.

Ans: The work done is **zero**.

Explanation:

Work done is given by

$$W = \text{force} \times \text{displacement}.$$

In one complete circular rotation, the object returns to its starting point. Therefore, its **displacement is zero**, even though it has travelled a distance.

Since displacement = 0,

$$W = F \times 0 = 0,$$

So, the work done in one complete circular path is **zero**.

29.

Ans: (a) 20 m/s²

Explanation:

Work $W = F \times s$

$$F = \frac{W}{s} = \frac{4000}{10} = 400 \text{ N}$$

$$\text{Acceleration } a = \frac{F}{m} = \frac{400}{20} = 20 \text{ m/s}^2.$$

30.

Ans: (b) 60 J

Explanation:

Kinetic energy of stone

$$\text{K.E.} = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.10 \times 30^2 = 45 \text{ J}$$

This 45 J is **75%** of the elastic band's potential energy (since 25% is wasted).

$$\text{P.E.} = \frac{45}{0.75} = 60 \text{ J}$$

31.

Ans: Dissipation of energy is the loss of useful energy when mechanical energy is changed into another form that cannot be used for work. This wasted energy is called **dissipated energy**.

32.

Ans: A **joule** is the work done when a force of **1 newton** moves a body through a distance of **1 metre** in the direction of the force.

The **SI unit** of work is **joule (J)**.

To relate SI and CGS units:

$$1 J = 1 N \times 1 m$$

We know,

$$1 N = 10^5 \text{ dyne}, \quad 1 m = 100 \text{ cm}$$

So,

$$1 J = 10^5 \times 100 = 10^7 \text{ dyne cm} = 10^7 \text{ erg}$$

Thus,

$1 J = 10^7 \text{ erg}$

33.

Ans: According to the law of conservation of energy, the **potential energy stored in the compressed spring** is equal to the **kinetic energy gained by the trolley**.

$$\begin{aligned} \text{K.E.} &= \frac{1}{2}mv^2 \\ &= \frac{1}{2} \times 0.5 \times 2^2 \\ &= 1 \text{ joule} \end{aligned}$$

Hence, the **potential energy of the spring = 1.0 J**.

34.

Ans: (a) 400 N

Explanation:

Work $W = F \times s$

$$F = \frac{W}{s} = \frac{8000}{20} = 400 \text{ N}$$

So, the required force is **400 N**.

35.

Ans: A kilowatt hour (kWh) is the amount of electrical energy used when an appliance of power **1 kilowatt** works for **1 hour**.

To convert it into joules:

$$1 \text{ kW} = 1000 \text{ W}$$

$$1 \text{ hour} = 3600 \text{ s}$$

$$1 \text{ kWh} = 1000 \times 3600 = 3.6 \times 10^6 \text{ J}$$

So,

$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$

36.

Ans:

Given:

Mass of the body, $m = 4 \text{ kg}$

Force applied, $F = 16 \text{ N}$

Initial velocity, $u = 0$

Time, $t = 10 \text{ s}$

Acceleration is:

$$a = \frac{F}{m} = \frac{16}{4} = 4 \text{ m/s}^2$$

Velocity after 10 s:

$$v = u + at = 0 + 4 \times 10 = 40 \text{ m/s}$$

Kinetic energy at the end of 10 s:

$$\text{K.E.} = \frac{1}{2}mv^2 = \frac{1}{2} \times 4 \times (40)^2 = 3200 \text{ J}$$

3200 J

37.

Ans: (c) negative work

Explanation:

The coolie lifts the load **upwards**, but the weight (force of gravity) acts **downwards**, opposite to the displacement.

Work done by a force opposite to displacement is **negative**.

So, work done **by the load (gravity)** is negative.

38.

Ans: (b) 10^7

Explanation:

The relation between Joule and Erg is:

$$1 \text{ J} = 10^7 \text{ erg}$$

Thus, option **(b)** is correct.

39. (i)

Ans: Correct option: **(b)** Kinetic energy of body *A* is greater than kinetic energy of body *B*.

Because for same momentum p

$$K = \frac{p^2}{2m}$$

So, kinetic energy is **inversely proportional** to mass. The lighter body *A* has **greater** kinetic energy.

39. (ii)

Ans: Correct option: **(c)** 2 : 5

Given

$$K_A : K_B = 5 : 2 \text{ and } K \propto \frac{1}{m} \text{ (same } p\text{)}$$

$$\frac{K_A}{K_B} = \frac{m_B}{m_A} \Rightarrow \frac{5}{2} = \frac{m_B}{m_A} \Rightarrow m_A : m_B = 2 : 5$$

So, the mass ratio $A : B = 2 : 5$

40.

Ans: (b) light to chemical

Explanation:

In photosynthesis, plants absorb **light energy** from the Sun and convert it into **chemical energy** stored in glucose.

41. (i)

Ans: (d) 20 J

Explanation:

Mass $m = 200 \text{ g} = 0.2 \text{ kg}$

Height $h = 10 \text{ m}$

Acceleration due to gravity $g = 10 \text{ m/s}^2$

$$\text{Potential energy} = mgh = 0.2 \times 10 \times 10 = 20 \text{ J}$$

Thus, the correct option is **(d)**.

41. (ii)

Ans: (c) 30 J

Explanation:

Total mechanical energy remains constant during free fall.

Initial energy at height 15 m:

$$E = mgh = 0.2 \times 10 \times 15 = 30 \text{ J}$$

So, just before hitting the ground, total mechanical energy is still **30 J**.

41. (iii)

Ans: (c) $10\sqrt{3}$ m/s

Explanation:

Just before hitting the ground, all 30 J of energy is kinetic:

$$\frac{1}{2}mv^2 = 30 \Rightarrow \frac{1}{2} \times 0.2 \times v^2 \Rightarrow 30 \Rightarrow v^2 = 300 \Rightarrow v = \sqrt{300} = 10\sqrt{3} \text{ m/s}$$

42.

Ans: Kinetic energy is the energy possessed by a body due to its motion.

If a body of mass m moves with velocity v , its kinetic energy is:

$$E_k = \frac{1}{2}mv^2$$

The **SI unit** of kinetic energy is the **joule (J)**.

Examples:

(i) A bullet fired from a gun can penetrate a target because it has kinetic energy.

(ii) A falling stone can break the glass top of a table due to its kinetic energy.

43.

Ans:

Given:

$$F = 150 \text{ N}, v = 10 \text{ m/s}$$

Power is given by:

$$P = Fv$$

$$P = 150 \times 10 = 1500 \text{ W}$$

So, the power exerted is **1500 W**.